OR Central Exercise

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Branch & Bound and Cutting Planes

5 June 2026

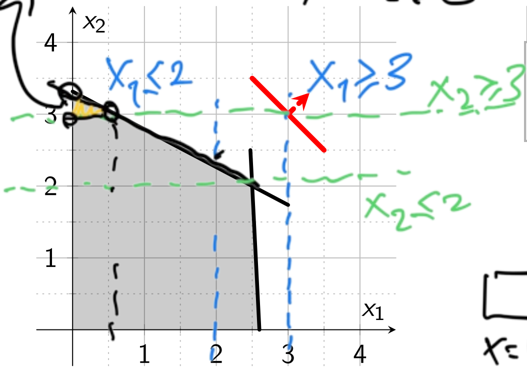
Question 1

$$
\begin{array}{llcl}
(P) & \max & x_1 + x_2 & \\
 & & 10x_1 + 19x_2 & \leq 63 \\
 & & 10x_1 + 0.5x_2 & \leq 26 \\
 & & x_1, x_2 & \cancel{\in \mathbb{N}_0} \geq 0
\end{array}
$$

Solve the integer linear program (P) using the branch-and-bound method in first-in-first-out *FIFO* order.

Pu

$x_1 \leq 2, x_2 \geq 3$



P_0
$x = (2.5, 2)$ $z = 4.5$ *branch w.r.t.* x_1

$x_1 \geq 3$ → P_2: infeasible

$x_1 \leq 2$ → P_1: $x = (2, 43/19)$, $z = 4,..$

$x_2 \leq 2$ → P_3: $x = (2, 2)$, $z = 4$, optimality

$x_2 \geq 3$ → P_4: $x = (0,.., 3)$, $z = 3,.. < 4$, suboptimality

Question 2

$$
\begin{array}{lrcl}
(P)\ \max & 4x_1 + 2x_2 & & \\
& 2x_1 + 3x_2 & \leq & 3 \\
& -2x_1 - 2x_2 & \geq & -5 \\
& x_1, x_2 & \cancel{\in} & \cancel{\mathbb{N}_0}
\end{array}
$$

$\longrightarrow \ldots \leq \ldots$

$\longrightarrow \geq 0$

add slack var.

part a)

Give the LP relaxation of (P) in canonical form.

Question 2

$$
\begin{array}{lrcl}
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 & 2x_1 + 3x_2 & \leq & 3 \\
 & -2x_1 - 2x_2 & \geq & -5 \\
 & x_1, x_2 & \in & \mathbb{N}_0
\end{array}
$$

part a)

Give the LP relaxation of (P) in canonical form.

$$
\begin{array}{lrrrrcl}
\max & 4x_1 & +2x_2 & & & & \\
 & 2x_1 & +3x_2 & +x_3 & & = & 3 \\
 & 2x_1 & +2x_2 & & +x_4 & = & 5 \\
 & & & x_1, x_2, & x_3, x_4 & \geq & 0.
\end{array}
$$

Question 2

$$\begin{array}{lrrrrcl}
\max & 4x_1 & +2x_2 & & & & \\
& 2x_1 & +3x_2 & +x_3 & & = & 3 \\
& 2x_1 & +2x_2 & & +x_4 & = & 5 \\
& & & x_1, x_2 & x_3, x_4 & \geq & 0.
\end{array}$$

part b)

Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

Question 2

$$\begin{array}{lcccccc} \max & 4x_1 & +2x_2 & & & & \\ & 2x_1 & +3x_2 & +x_3 & & = & 3 \\ & 2x_1 & +2x_2 & & +x_4 & = & 5 \\ & & & x_1, x_2 & x_3, x_4 & \geq & 0. \end{array}$$

part b)

Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

	x_1	x_2	x_3	x_4	RHS
z	-4	-2	0	0	0
x_3	2	3	1	0	3
x_4	2	2	0	1	5

Question 2

$$\begin{array}{lrrrrcl}
\max & 4x_1 & +2x_2 & & & & \\
& 2x_1 & +3x_2 & +x_3 & & = & 3 \\
& 2x_1 & +2x_2 & & +x_4 & = & 5 \\
& & & x_1, x_2 & x_3, x_4 & \geq & 0.
\end{array}$$

	x_1	x_2	x_3	x_4	RHS
z	-4	-2	0	0	0
x_3	2	3	1	0	3
x_4	2	2	0	1	5

$\Longrightarrow \quad x_1 \leftrightarrow x_3$

part b)

Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

Question 2

$$\begin{array}{lrrrrcl} \max & 4x_1 & +2x_2 & & & & \\ & 2x_1 & +3x_2 & +x_3 & & = & 3 \\ & 2x_1 & +2x_2 & & +x_4 & = & 5 \\ & & & x_1, x_2 & x_3, x_4 & \geq & 0. \end{array}$$

part b)

Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

	x_1	x_2	x_3	x_4	RHS
z	-4	-2	0	0	0
x_3	2	3	1	0	3
x_4	2	2	0	1	5

$\implies x_1 \leftrightarrow x_3$

	x_1	x_2	x_3	x_4	RHS
z	0	4	2	0	6
x_1	1	1.5	0.5	0	1.5
x_4	0	-1	-1	1	2

$$
\begin{array}{lllllll}
\max & 4x_1 & +2x_2 & & & & \\
 & 2x_1 & +3x_2 & +x_3 & & = & 3 \\
 & 2x_1 & +2x_2 & & +x_4 & = & 5 \\
 & & & x_1, x_2 & x_3, x_4 & \geq & 0.
\end{array}
$$

part b)

Solve the LP relaxation using the simplex method and give the optimal solution and the optimal objective function value.

	x_1	x_2	x_3	x_4	RHS
z	-4	-2	0	0	0
x_3	2	3	1	0	3
x_4	2	2	0	1	5

	x_1	x_2	x_3	x_4	RHS
z	0	4	2	0	6
x_1	1	1.5	0.5	0	1.5
x_4	0	-1	-1	1	2

$\implies x_1 \leftrightarrow x_3$

The optimal solution is thus $(x_1, x_2) = (1.5, 0)$ with $Z = 6$.

	x_1	x_2	x_3	x_4	RHS
z	0	4	2	0	6
x_1	1	1.5	0.5	0	1.5
x_4	0	−1	−1	1	2

part c)

Give the inequality of a possible Gomory cut with respect to the determined optimal tableau.

$x_1 = 1.5$

$x_2 = 0$

x_1 should be integer, but it is not.

$$x_1 + 1.5x_2 + 0.5x_3 = 1.5$$

integer

fractionals

$$x_1 + x_2 - 1 = 0.5 - 0.5x_2 - 0.5x_3$$

cuts off

$$\Rightarrow x_1 + x_2 - 1 \leq 0 \ \& \ 1 - x_2 - x_3 \leq 0$$

Question 2

	x_1	x_2	x_3	x_4	RHS
z	0	4	2	0	6
x_1	1	1.5	0.5	0	1.5
x_4	0	-1	-1	1	2

part c)

Give the inequality of a possible Gomory cut with respect to the determined optimal tableau.

There is only one base variable with a non-integer value: $x_1 = 1.5$

Split it into an integer part and a remainder part:

The Gomory cut:

Question 3

$$\max\ z = x_1 + x_2$$
$$\text{s.t.}\ \ x_1 + 2x_2 + s_1 = 5$$
$$3x_1 + x_2 + s_2 = 6$$
$$x_1 + s_3 = 2.4$$
$$x_2 + s_4 = 3.8$$
$$s_1, s_2, s_3, s_4 \geq 0$$
$$x_1, x_2 \in \mathbb{N}_0$$

	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	$2/5$	$1/5$	0	0	$16/5$
x_1	1	0	$-1/5$	$2/5$	0	0	$7/5$
x_2	0	1	$3/5$	$-1/5$	0	0	$9/5$
s_3	0	0	$1/5$	$-2/5$	1	0	1
s_4	0	0	$-3/5$	$1/5$	0	1	2

part a)

Identify the basic variables that can be used to generate Gomory fractional cuts, and explain why they are eligible.

x_1 & x_2 because x_1, x_2 supposed to $\in \mathbb{N}_0$ but they are not.

Question 3

$$\max z = x_1 + x_2$$
$$\text{s.t. } x_1 + 2x_2 + s_1 = 5$$
$$3x_1 + x_2 + s_2 = 6$$
$$x_1 + s_3 = 2.4$$
$$x_2 + s_4 = 3.8$$
$$s_1, s_2, s_3, s_4 \geq 0$$
$$x_1, x_2 \in \mathbb{N}_0$$

	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	2/5	1/5	0	0	16/5
x_1	1	0	−1/5	2/5	0	0	7/5
x_2	0	1	3/5	−1/5	0	0	9/5
s_3	0	0	1/5	−2/5	1	0	1
s_4	0	0	−3/5	1/5	0	1	2

cut off

part b)

The company tests both Gomory cuts. Derive **all** valid representations of **each** Gomory cut from the tableau above.

$$x_2 + \frac{3}{5}s_1 - \frac{1}{5}s_2 = \frac{9}{5}$$

$$\to x_2 - s_2 - 1 \leq 0$$

$$\to 4 - 3s_1 - 4s_2 \leq 0$$

x_1

$$\Rightarrow x_1 - s_1 - 1 \leq 0 \quad \& \quad 2 - 4s_1 - 2s_2 \leq 0$$

Question 3

$$\max\ z = x_1 + x_2 \leq 3$$

$$\text{s.t.}\quad x_1 + 2x_2 + s_1 = 5$$
$$3x_1 + x_2 + s_2 = 6$$
$$x_1 + s_3 = 2.4$$
$$x_2 + s_4 = 3.8$$
$$s_1, s_2, s_3, s_4 \geq 0$$
$$x_1, x_2 \in \mathbb{N}_0$$

$$\Rightarrow s_1 = 5 - x_1 - 2x_2$$

	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	$2/5$	$1/5$	0	0	$16/5$
x_1	1	0	$-1/5$	$2/5$	0	0	$7/5$
x_2	0	1	$3/5$	$-1/5$	0	0	$9/5$
s_3	0	0	$1/5$	$-2/5$	1	0	1
s_4	0	0	$-3/5$	$1/5$	0	1	2

part c)

Simplify the Gomory cut obtained from the x_1-row so that it can be directly incorporated into the integer program (use the form without slack variables).

$$x_1 - \cancel{s_1} - 1 \leq 0$$
$$+\ x_1 + 2x_2 + \cancel{s_1} = 5$$

(1) $2x_1 + 2x_2 - 1 \leq 5 \Rightarrow 2x_1 + 2x_2 \leq 6$

(2) $x_1 - (5 - x_1 - 2x_2) - 1 \leq 0$

$$2x_1 - 6 + 2x_2 \leq 0$$

Question 3

$$\max z = x_1 + x_2 \quad (1)$$
$$\text{s.t. } x_1 + 2x_2 + s_1 = 5 \quad (2)$$
$$3x_1 + x_2 + s_2 = 6 \quad (3)$$
$$x_1 + s_3 = 2.4 \quad (4)$$
$$x_2 + s_4 = 3.8 \quad (5)$$
$$s_1, s_2, s_3, s_4 \geq 0 \quad (6)$$
$$x_1, x_2 \in \mathbb{N}_0 \quad (7)$$

	x_1	x_2	s_1	s_2	s_3	s_4	
z	0	0	2/5	1/5	0	0	16/5
x_1	1	0	−1/5	2/5	0	0	7/5
x_2	0	1	3/5	−1/5	0	0	9/5
s_3	0	0	1/5	−2/5	1	0	1
s_4	0	0	−3/5	1/5	0	1	2

part d)

Formulate the updated integer program, including the first Gomory cut. You may reference earlier equations by their number (e.g., (1)) instead of rewriting them.

max (1)
s.t. (2)−(7)

add the cut: $x_1 + x_2 \leq 3$

$x_1 + x_2 + s_5 = 3$
$s_5 \geq 0$